CHUNKING MEMORY VS RECALL

TITLE:

Design a User Interface where users recall visual elements and evaluate the effect of chunking on memory performance.

AIM:

To design and develop a prototype of a memory recall task using chunked visual elements and to evaluate the impact of chunking on users' memory performance.

SOFTWARE USED:

• Figma (UI Design Tool).

THEORY:

Memory recall refers to an individual's ability to retrieve previously viewed information from short-term memory. Cognitive psychology indicates that humans can store only a limited amount of information within short-term memory at a given time.

Chunking is a memory enhancement strategy in which information is organized into meaningful groups or units. According to Miller's Law (the 7 ± 2 rule), individuals can typically retain about five to nine items in short-term memory. Organizing information into groups enhances recall efficiency by minimizing cognitive load.

This experiment aims to evaluate the influence of chunked visual elements on users' memory recall performance.

DESIGN DESCRIPTION:

A. Home Screen

The home screen contains:
• Title: "Memory Recall Task"
• Instructions explaining the task
• Viewing duration (5 seconds)
• Start button to begin the experiment

Purpose:

To provide clear instructions before beginning the memory task.

B. Chunking Phase

The chunking phase presents the following elements:

• Multiple automobile brand logos displayed as visual icons

• Logos organized into distinct visual groups to support chunk-based learning

• A countdown timer set to 5 seconds for observation

Purpose:

To allow users to observe grouped visual elements and encode the information into short-term memory using the chunking technique, thereby improving recall efficiency.

00:03

00:02

00:01

C. Recall Phase

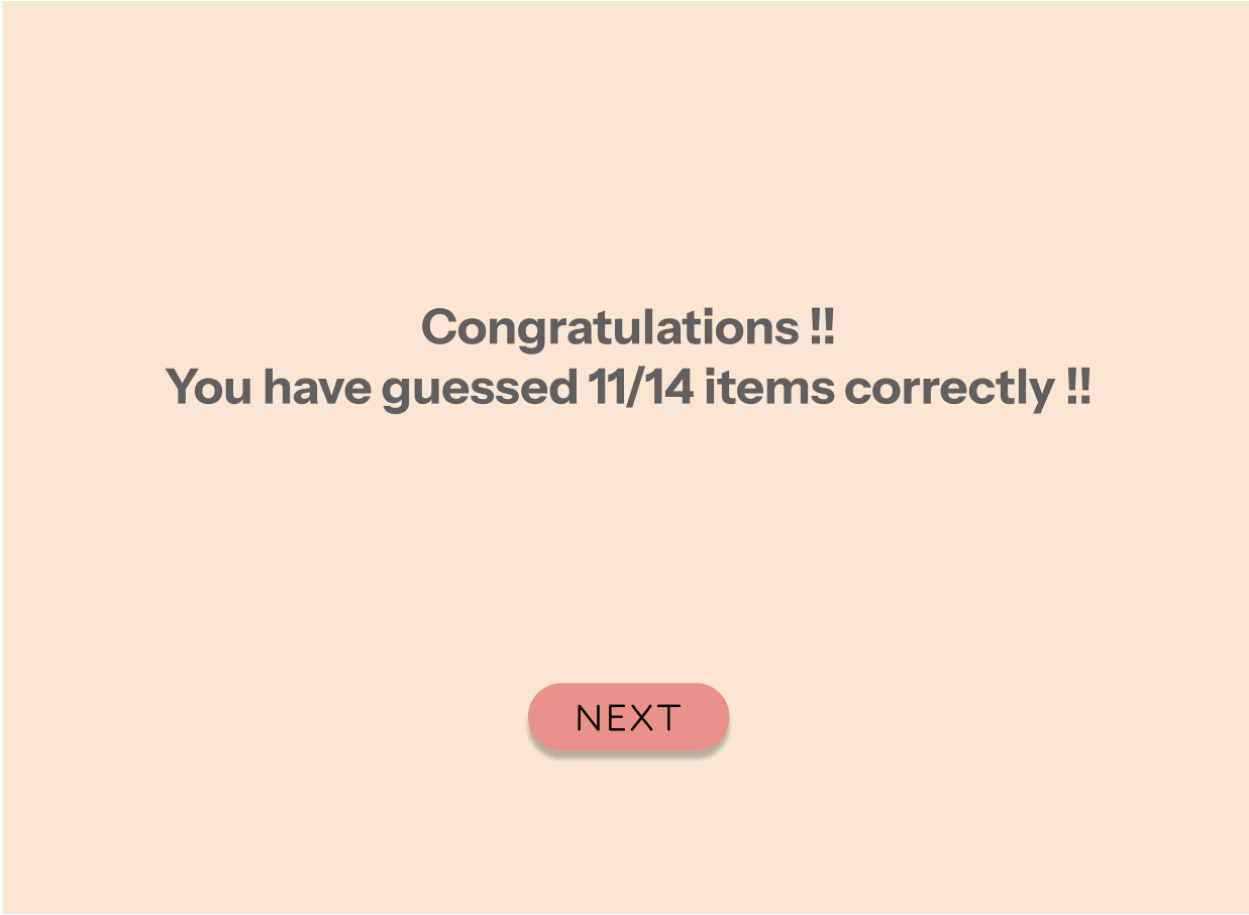
The recall screen includes:
- Text prompting users to recall items
- Input fields to enter remembered items
- Submit button

Purpose: To measure recall accuracy after the viewing phase.

D. Result Screen

The result screen displays:
- Feedback message
- Score (e.g., you have guessed 11/14 items correctly)
- Next button

Purpose: To provide performance feedback to the user.



PROCEDURE:

- Developed a Home Screen containing instructions and a Start button to initiate the task.
- Designed the Chunking Phase to display visually grouped icons.
- Implemented a prototype delay of 5 seconds to simulate the observation period.
- Created a Recall Phase with input fields to capture user responses.
- Added a Submit button connected to the Result Screen.
- Displayed the final score based on the accuracy of user recall.

OBSERVATION:

Users demonstrated improved recall when the icons were presented in visually grouped formats. The use of chunking minimized confusion and enhanced recognition patterns.

RESULT:

The experiment indicates that chunking significantly enhances short-term memory recall performance.

CONCLUSION:

Chunking improves memory retention by organizing information into structured and meaningful groups. The findings confirm that visually grouped elements lead to higher recall accuracy compared to unstructured presentations.